

A Comprehensive Survey on Transparency and Traceability in Agricultural Supply Chain Management Using Customized Smart Contracts and Blockchain Technologies

Abstract: Transparency and traceability take the spotlight in agricultural supply chain management, driven by food safety and fraud, sustainability, and regulatory concerns. Fractured data, siloed, and visibility-less record keeping, and lack of trust among farmers, processors, distributor, retailer, and consumer, describe most traditional agri supply chain. These drawbacks impede the supply chain real-time tracking, accountability, and decision making. In the last few years, the decentralized, immutable, and adaptable data management of blockchain technology has captured interest in solving supply chain management problems. The additional layer of custom smart contracts further promotes the automation, trust of supply chain stakeholders and controlling the flow of supply chain operations. This paper attempts to be the first to collate the scattered literature on the use of blockchain technology in the agricultural supply chain management within the scope of customized smart contracts. It analyzes and classifies the literature based on consensus algorithms, blockchain structures, smart contract, data capture (IoT and sensing tech.), data traceability, and maintained records. It also provides an in-depth analysis of the major application areas, such as crop production, livestock management, food processing, food logistics, and retail distribution. The survey also describes performance evaluation indicators which include scalability, latency, throughput, security, cost efficiency, and interoperability. The survey also describes open issues and the challenges of adopting blockchain in agricultural supply chains, including scalability issues, the vulnerability of smart contracts, integration problems, privacy/data issues, and regulation. Some challenges include the vulnerability of smart contracts and integration problems, as well as the regulation of privacy/data issues. Other challenges include the vulnerability of smart contracts and integration issues, as well as regulatory concerns on data privacy. The survey also identifies developing phenomena and potential future studies, such as blended blockchain solutions, optimization of off-chain storage, automation of contracts through artificial intelligence, and smart contracts that comply with specific policies. The survey provides a comparative study and taxonomy and aims to provide a reference to researchers, practitioners, and policymakers in setting up agricultural supply chains that are transparent, secure, and easily traceable, using blockchain and smart contracts.

Keywords: Blockchain, Supply Chain Management, IoT Integration, Smart Contracts, Data Security, Traceability, Cybersecurity, Digital Transformation

Introduction

A supply chain, or SCM system, includes industries like farming, agriculture, and food production. Factors like food safety, food quality, food sustainability, and the availability of food all over the world, depend largely on Agricultural supply chain management (ASCM). A supply chain consisting of agriculture and farming typically is made up of many entities. A few of the more prominent entities would be farmers, processors, aggregators, transporters, wholesalers, retailers, and the final consumers. Such a supply chain is of prime importance. However, various problems continue to cause issues in the supply chain for farming and agriculture. A few issues are gaps in trust, systems, and transparency pertaining to fragmentation, lack of systems, shoddy traceability, and some more issues. Another problem and challenge to the farming supply chain is food safety. There are various types of food safety problems such as food fraud, food adulteration, counterfeit

foods, delayed product recalls, and more. Such issues and problems will greatly impact popular trust and compliance with the various global food safety regulations. The food and safety global regulations are issued and monitored by The World Food Organization. Valid, accurate, and current supply chain systems are highly lacking in the supply chain industry. Valid, accurate, and current supply chain systems are lacking and very few exist. Most existing systems are still using outdated systems to trace and document food and food products. Invisible systems are seen to be present and are believed to be lacking to monitor the systems in an automated and fair basis. The systems are fair and automated but are seen to be lacking. A technology system will work in a fair manner. The below figure 1 shows the conceptual blockchain framework in IoT with agriculture. With the blockchain, wise supply chain collaboration and facilities can be recorded in a wise, chain, and centrally authoritative dependent fashion, and the backward end traceability of agricultural chain products can be achieved. This survey attempts to provide a consolidated framework of knowledge and establish a definable research path to assist system designers, policymakers, and researchers in constructing systems that are high functioning, transparent, and trustworthy for agricultural supply chains.

Literature review

The primary purpose of this section is the focal exegesis of the salient trends, technology, and frameworks of research, esp. concerning data collection and its related applications, IoT, and smart contracts in blockchain-enabled supply chain. It is the first critical stage of setting an infrastructure for the metric comparative analysis stages.

Blockchain Foundations for Transparency and Traceability in Supply Chains

The potential of blockchain technology as a means of providing greater transparency, traceability, and trust in supply chains has been widely documented. Many of the initial studies in this area, focusing on almost all supply chains, point out that the supply chains are built around central databases and rely on intermediary third parties, which creates a hostile environment in which data can be manipulated, there are no real-time updates, and audits are impossible [1]. A supply chain using blockchain technology can provide a fully decentralized supply chain because a blockchain acts as a completely immutable, verifiable, and shareable ledger for a defined set of users and can provide traceability for all transactions. This is particularly applicable for supply chains in the agri-food industry; in supply chains where the quality of the product itself is a concern (i.e. the product can spoil, be contaminated, or be misrepresented), the use of blockchain technology in supply chains is most economically beneficial. For example, Aggarwal et al. explain that, in addition to helping achieve compliance with safety standards, a framework that utilizes blockchain in traceability systems for a food supply chain enhances the food provenance by allowing all participants to trace a food product from the farm to the consumer [2]. Such systems eliminate information asymmetry by providing traceable certified records to the consumers. Transparent records keep the consumers confident in the supply chain and the system as a whole. Blockchain supply chains that are regulated by government authorities have the same cost-benefit effect; in the supply chain of a food-based product, blockchain is used to prevent counterfeit medicines from entering the supply chain and to verify the authenticity of the products in the market [3]. Looking at the overall picture, blockchain is beneficial for supply chains in all sectors in terms of improving and maintaining security, preventing unauthorized users from modifying data, and ensuring accountability among users [4].

Despite this, these studies also admit the challenges blockchain adoption raises including complexities in integration, scalability and governance issues. The gaps in supply chains

are further compounded by the advanced persistent threats that take advantage of the gaps in digital reliance, thus showing the need for resilient architectures in blockchain [5]. The most recent research pushes the boundaries of blockchain adoption beyond the recording of data to the assimilation of digital twins and Industry 4.0 frameworks. The digital twin - blockchain assimilation facilitates real time synchronization of the physical and digital representations of the supply chain which in turns enables predictive monitoring and operational optimization [6]. Supplementary studies stress that the autonomous decision making process of blockchain could be further improved through the incorporation of artificial intelligence (AI) and Internet of Everything (IoE) technologies [7]. The issue of efficiency continues to raise concern for most conventional blockchain implementations in terms of latency and throughput. The mechanism of traceability for blockchain in large supply chains proposed by Wu et al. focuses on the optimization of consensus and data structures to enhance efficiency [8]. Moreover, studies centred on IoT emphasizes that blockchain provides the needed integration of decentralized collaboration among the various actors in the logistics chain while ensuring the integrity of data generated by the sensors [9]. Empirical analyses also confirm that the integration of IoT and blockchain technologies accelerates the growth of supply chains through improved coordination, traceability and trust [10].

Table 1.: Blockchain Foundations for Supply Chain Transparency and Traceability

Author(s) & Citation	Application Domain	Methodology / Approach	Key Contribution	Limitations
Dudeczyk et al. [1]	Global supply chains	Comprehensive survey of blockchain architectures	Identified transparency and traceability benefits across sectors	Lacks domain specific agricultural evaluation
Aggarwal et al. [2]	Agri-food supply chain	Blockchain based traceability framework	Improved end-to-end food provenance and trust	Scalability concerns in large deployments
Mangala et al. [3]	Pharmaceutical supply chain	Blockchain integrated with IoT cloud	Secure authentication and anti-counterfeiting	High system complexity
Islam [4]	Supply chain security	Systematic survey	Classified blockchain security mechanisms	Limited performance analysis
Wu et al. [8]	Transportation supply chain	High efficiency blockchain traceability	Reduced latency and storage overhead	Limited real world validation

IoT, Digital Twins, and Infrastructure Integration for Traceability

The literature's second phase examines the cutting-edge integration of blockchain technology with the Internet of Things (IoT), digital twins, and cyber–physical systems for real-time, data-driven traceability in supply chain management. Research emphasizes that, while blockchain guarantees the immutability of data, IoT devices are the primary data creators, recording the environmental, operational, and logistical aspects of the supply chain [11]. The combination of blockchain and IoT enhances the trustworthiness of data, minimizes the need for human involvement, and improves transparency. In contrast, supply chains that use IoT systems present new threats to security and challenges to scalability.

Security frameworks based on blockchain technology have been created to secure IoT networks from unauthorized access and modification of data [12]. System reliability is further undermined by software supply chain vulnerabilities; dependencies and third party components are added to provide concealed attack vectors [13]. The analyses of cybersecurity suggest that supply chain components be protected from technological and human adversaries to ensure uninterrupted operations [14], [15]. The noted studies have documented the economic and reputational consequences of ransomware attacks on digital supply chains and have pointed out that ransomware attacks are the most dangerous on digital supply chains [16], [17]. The food supply chain is the most affected by these attacks because of its operational reliance on flow and time sensitivity of the products [18]. The categorization of underlying components of these threats and their countermeasures, such as decentralized access control, encrypted data sharing, and anomaly detection, are documented in comprehensive surveys [19]. Approaches that integrate advanced detection mechanisms based on adaptive machine learning models to identify zero-day exploits and other malicious activities within digital supply chain infrastructures have been proposed [20]. These approaches combine with the preventive security mechanisms of blockchain technology to provide advanced proactive threat detection.

Table 2.: IoT, Digital Twins, and Infrastructure Integration

Author(s) & Citation	Technology Focus	Integration Model	Major Findings	Challenges Identified
Roumeliotis et al. [6]	Blockchain + Digital Twins	Industry 4.0 framework	Enabled predictive supply chain monitoring	Interoperability complexity
Shoomal et al. [9]	IoT-enabled SCM	IoT integration survey	Enhanced resilience and efficiency	Data privacy risks
Singh & Dadhich [10]	Blockchain + IoT	Conceptual adoption model	Improved coordination and visibility	Lack of implementation
Maurya et al. [12]	IoT security	Blockchain driven security model	Strengthened IoT data protection	Computational overhead

Mohamed et al. [20]	Cybersecurity	ML-based exploit detection	Proactive attack detection	Requires high quality datasets
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Smart Contracts and Automated Supply Chain Governance

The ability to automate transactions and execute them based on predetermined conditions has the potential to greatly enhance the value of blockchain-based supply chain systems, particularly in smart contracts. Research shows that supply chain software can be protected through techniques like machine learning to understand behaviors that cause problems before they affect smart contracts [21], [22]. The study of MITM attacks highlights the possible problems with unsecured channels of communication with smart contract systems [23]. Social engineering attacks on developers are an example of a more human threat that smart contract systems face [24?]. Proposed methods of securing supply chain data involve a blockchain with an encryption schema that protects sensitive data and secure storage [25]. The trust in smart contracts is improved by blockchain-integrated frameworks that ensure the data from the IoT remains unaltered [26, 27, 28]. From the viewpoint of operational procedures, the adoption of the blockchain significantly changes the nature of trust among supply chain partners, especially concerning the organic and sustainable food systems [29]. The secure data sharing means of port and logistics supply chains shows how smart contracts enable the elimination of access control and the retention of privacy [30]. The real-world use case shows how smart contracts handle the automation of IoT supply chains in procurement, of tracking inventory, and of settling payments [31, 32]. Theoretical and empirical work has established that smart contracts that are tailored to contexts are more efficient by encoding rules that are specific to a given domain in a way that is transparent and that minimizes the possibility of controversy. [33]. Advance cash credit payments are an example of the financial mechanisms based on smart contracts that enhance the liquidity of the supply chains [34]. Projects that use Ethereum have proven the practicality of smart the security and gas cost considerations of contracting at scale [35].

Table 3. : Smart Contracts and Automated Supply Chain Operations

Author(s) & Citation	Smart Contract Role	Platform / Tool	Contribution	Limitations
Naqvi et al. [31]	SCM automation	Blockchain + IoT	Demonstrated smart contract workflow	Security not fully evaluated
Sakib [33]	Trust enforcement	Blockchain based contracts	Improved transparency and efficiency	Legal enforceability unclear
Raj et al. [34]	Financial transactions	Blockchain smart contracts	Automated payments and procurement	Limited scalability analysis
Muthamilselv	Ethereum	Solidity smart	Practical Ethereum	Gas cost

an et al. [35]	SCM	contracts	deployment	overhead
Yigit & Dag [36]	Process optimization	Ethereum network	Improved operational efficiency	Smart contract vulnerability risk

Cryptography, Security Threats, and Resilience in Blockchain Supply Chains

Security and resilience is the main theme of the Phase IV literature. Ethereum-based smart contract studies suggest that coding optimizations should always incorporate secure coding techniques to avoid exploits and inefficiencies [36]. Digital signatures and other cryptographic techniques support authentication and non-repudiation within blockchain systems [37], [38]. The integration of lightweight cryptography and blockchain has been proposed to secure sensitive information while optimizing performance [39]. Model frameworks that incorporate blockchain to provide privacy and transparency for IoT enabled supply chain systems have demonstrated improved data protection and compliance with regulations [40]. Surveys of distributed ledger technology also note that scalability and fault tolerance are important design attributes [41], [42]. Reviews suggest that implementing blockchain in supply chain systems improves security and transparency, but it needs to be managed with care. Stakeholders need to be coordinated as well [43]. Blockchain technology is also used in algorithmic systems designed to detect counterfeit products and assure quality [44]. Frameworks on counterfeiting traceability further capture the essence of blockchain in product authentication [45]. Literature reviews on supply chain blockchain applications also capture the benefits and drawbacks of the implementations [46], [47]. Cutting-edge literature reviews also mention the unresolved security issues of smart contract vulnerabilities and privacy leakage [48]. Studies on the convergence of blockchain and IoT emphasize the need for secure validated transactions [49]. Research on the provenance of data demonstrates that blockchain technology also enables sustainable and accountable supply chains [50].

Table 4.: Security Threats, Cryptography, and Resilience

Author(s) & Citation	Security Focus	Technique Used	Key Findings	Research Gap
Hammi & Zeadally [15]	Software supply chain security	Threat taxonomy	Identified major attack vectors	No blockchain mitigation
Manning & Kowalsk [18]	Ransomware in food SCM	Risk analysis	Highlighted sector specific threats	Lacks technical countermeasures
Zhou et al. [21]	Malware detection	LSTM + SVM	High detection accuracy	Model complexity
Lin et al. [25]	Data confidentiality	Blockchain encryption	Secure decentralized storage	Key management complexity

Agarwal et al. [48]	Blockchain security	Comprehensive survey	Identified emerging attack surfaces	Needs quantitative evaluation
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Sustainability, Energy Efficiency, and Future Directions

The concluding phase tackles sustainability, governance, and potential avenues for future research. More than commercial supply chains, blockchain has also been deployed for humanitarian and philanthropic purposes, illustrating its social value [51]. A systematic review highlights the lack of research focused on blockchain-based food supply chains in developing countries [52]. The combination of blockchain and IoT frameworks strengthens food security through enhanced monitoring and traceability transparency [53], [54]. Certain game-theory models examine the incentives for blockchain adoption in agricultural supply chains and the incentivizing role of the government [55]. The intersection of the blockchain and circular economy advocates for sustainability and optimal resource utilization [56]. The digital transformation of the agricultural sector frameworks identifies blockchain as a primary agent of transformative change for creating resilient food supply chains [57]. The studies of energy consumption of blockchain technologies and the subsequent research on sustainable and less energy-consuming consensus mechanisms [58], [59], [60] are a result of the concerns of the advocates of sustainability of blockchain technology and the energy consumption that blockchain technologies are creating. This survey anticipates the future agricultural supply chains to be anchored in flexible governance frameworks, smart contracts, energy-efficient blockchains, AI-based security, and adaptive policy frameworks.

Table 5.: Sustainability, Governance, and Future Directions

Author(s) & Citation	Focus Area		Key Contribution	Limitation
Ellahi et al. [52]	Food SCM sustainability	Systematic review	Identified unexplored blockchain opportunities	Limited empirical data
Gupta & Shankar	Food security	Blockchain traceability	Improved decision transparency	Adoption barriers

[54]		system		
Hamidoğlu et al. [55]	Adoption incentives	Game theoretic model	Encouraged blockchain uptake	Assumes rational stakeholders
Sharma et al. [56]	Circular economy	Empirical analysis	Enabled sustainable agricultural SCM	Region-specific findings

Asif & Hassan [59]	Energy efficiency	Consensus comparison	Highlighted PoW vs PoS trade-offs	Does not consider hybrid models
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Multiple research papers focus on automated contract execution and the secure storage of data. Lin et al. [25] applied smart contracts and decentralised data storage to improve data confidentiality and integrity, although this still comes with the challenge of computational overhead. Naqvi et al. [31] addressed the automation of supply chain transactions using smart contracts initiated by the Internet of Things (IoT), which improves efficiency, although it still has scale problems it is similar like figure 2.



Figure 2: blockchain IoT integration for real time traceability

Additional contributions center around insights on adoption, privacy, and security. Yigit and Dag [36] exemplify process automation with Ethereum smart contracts and a Solidity based implementation, whereas Qatbi and Rathinam [40] tackle data privacy on blockchains via a blockchain access control mechanism.

Research Gaps and Open Challenges

Although there is increased academic literature on blockchain technology adoption in the traceability and transparency in agricultural supply chain management, there are still gaps that have yet to be addressed that are critical and pose open problems. The gaps in the literature provide an overview of the lack of research in the technical, organizational, security, scalability, and sustainability frameworks.

1. Scalability and Performance Bottlenecks: One of the biggest issues with agricultural supply chains based on blockchains is their implementation on a large scale. Most research cites proof-of-concept frameworks and few address these large-scale deployments. These large-scale deployments would address thousands of participants in the agricultural supply chain, high-volume and fast transactions, and streaming data from IoT devices.

2. Interoperability and Standardization Issues: Demanding different participants and older systems, Agricultural Supply Chains (ASC) have always been heterogeneous. Research on the interoperability of the different blockchain networks, the internet of things (IoT) platforms, and enterprise resource planning (ERP) systems is still sparse.

3. Data Privacy and Confidentiality Concerns: Even though blockchain improves the transparency of information, the transparency itself can be problematic when it comes to sensitive business and personal information.

4. Smart Contract Security and Reliability: Smart contracts are at the center of automating operations within the agricultural supply chain; however, worries remain regarding their reliability and security. Many existing implementations use static and hard-coded smart contracts which are hard to modify post-deployment. Financial losses and system errors are still possible because of logic errors, re-entrancy attacks, and insufficient access control.

5. Trust in Off-Chain Data and Oracle Mechanisms: Blockchain technology is always incorporated with technology which has been linked with external data sources like IoT sensors and third-party oracles to capture real-time data. Though blockchain guarantees the integrity of data captured, it does not guarantee the truthfulness of the data.

Conclusion

This survey explores the potential of blockchain technology and tailored smart contracts to reinforce transparency, traceability, and trust throughout supply chains, particularly in the agricultural sector. It highlights the main contributions to blockchain technology and agricultural supply chain literature by reviewing 60 studies and focuses on 15 of them for comparative analysis. The literature reveals that blockchain-based supply chain frameworks help to reduce untrustworthy and unproductive supply chain practices through enhanced record-keeping, real-time monitoring, and automated processing. Furthermore, the integration of blockchain and the IoT fosters supply chain transparency and offers stakeholders the ability to track and trace a product's origin, quality, and movement in real-time. Significant literature is available on the persistent challenges and unexplored barriers to the adoption of blockchain technologies and smart contracts in supply chains, particularly in large-scale agriculture. These include scalability, interoperability, and energy efficiency. Other unexplored blockchain in supply chain literature includes the adoption and enforcement of smart contracts, the non-alignment of international trade laws surrounding blockchain, and the lack of compliance with smart contracts. Researchers suggest that there is a need for formally articulated and uniformly established processes to benchmark and assess blockchain in supply chains systems. The comparative literature research illustrates that operational smart contracts and IoT integration are more efficient than merely offering theory-based conceptual frameworks. Nevertheless, the accompanying implementations almost always run into hurdles that demand the exercising of heavy computational processes that rely on complex cryptography tools needed to protect sensitive information. Lightweight blockchain structures, hybrid consensus, and smart contract optimizations powered by artificial intelligence should be the focus of future studies. Innovation and trust across the agricultural sector can be promoted by closing the existing research gaps which can help address blockchain technology and tailored smart contracts, and their possible impacts on agricultural supply chains. These technologies can help transform the management of supply chains by providing automated processes that are secure and transparent while enhancing the trust of stakeholders and sustainable resilient food systems. It will help provide a set of applicable, scalable, and cross the border efficient innovations.

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